

Multiple Choice (10 marks)

1. A sender must not be able to deny sending a message that was sent, is known as
 - (a) Message Nonrepudiation
 - (b) Message Integrity
 - (c) Message Confidentiality
 - (d) Message Sending
2. Of the following page-replacement policies, which is guaranteed to incur the minimum number of page faults?
 - (a) Replace the page whose next reference will be the longest time in the future.
 - (b) Replace the page whose next reference will be the shortest time in the future.
 - (c) Replace the page whose most recent reference was the shortest time in the past.
 - (d) Replace the page whose most recent reference was the longest time in the past.
3. Which of the following deals with network intrusion detection and real-time traffic analysis?
 - (a) John the Ripper
 - (b) L0phtCrack
 - (c) Snort
 - (d) Nessus
4. Failed sessions allow brute-force attacks on access credentials. This type of attacks are done in which layer of the OSI model?
 - (a) Physical layer
 - (b) Data-link Layer
 - (c) Session layer
 - (d) Presentation layer
5. Which form of encryption does WPA use?
 - (a) Shared key
 - (b) LEAP
 - (c) TKIP
 - (d) AES

True / False (10 marks)

Answer True or False

6. True or False: Message authentication is a service that ensures message integrity by verifying that the message has not been altered during transmission.
7. True or False: The stack is primarily used for storing local variables within a function's execution context.
8. True or False: Bubble sort achieves an average-case time complexity of $O(n \log n)$ in most scenarios.
9. True or False: The authentication process between a wireless user and an access point is known as the 4-way handshake.
10. True or False: The string 'xyz' can be derived from the grammar, as it follows the production rules provided.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find sum and average of first n natural numbers.
12. Write a python function to convert the given string to lower case.

End of Paper